Name: KOMAL MEHMOOD

SAP ID: 68618

DSA - ASSIGNMENT - 2

21 - feb - 2026

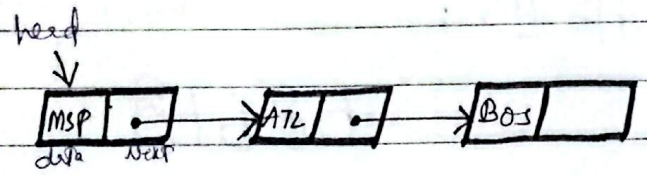
CamScanner

Date: Day:

1. Singly Linked List

A singly linked list is a data structure made up of nodes. Each node has two parts:

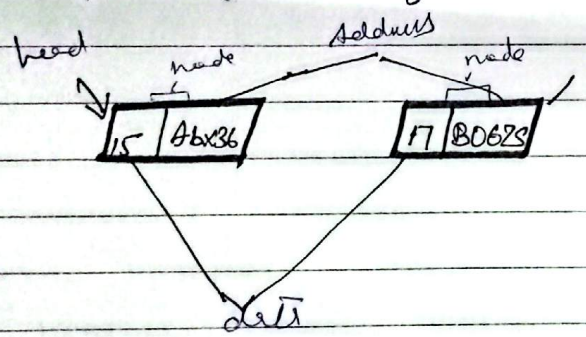
1. stores data.
2. stores the address of next node.



Node:

A node is the basic unit of a linked list. it contains:

- Data (value stored)
- Next pointer (address of the next node)

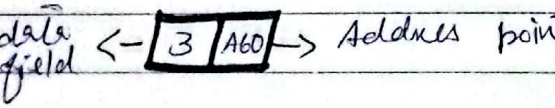


Data Field:

The data field stores the actual value in the node, such as 10, 20 or 30.

Next pointer:

The next pointer stores the address of the next node in the list



Header Pointer:

The header pointer stores the address

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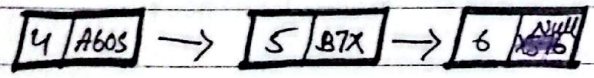
of the node of the linked list. It is used to access the entire list.



Part A: Insertion Operations:

Pseudocode:

```
Create New node N
N.data = 10
N.next = head
head = N
```

Dry run:

Head -> 20 -> 30 -> 40 -> Null

After insertion:

Head -> 10 -> 20 -> 30 -> 40 -> Null

Insertion at the End

Examples

Insert 50 into

10 -> 20 -> 30

Pseudocode:

```
Create new node N
N.data = 50
N.next = Null
temp = head
while Temp.next ≠ Null
        temp = temp.next
temp.next = N
```

Dry Run:

10 -> 20 -> 30 -> Null

Final List:

10 -> 20 -> 30 -> 50 -> Null

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Insertion at a specific position

Example: insert at 25 position into 3.

10 -> 20 -> 30 -> 40

Pseudocode:

```
create new node N
N.data = 25
temp = head
for i = 1 to position - 2
    temp = temp.next
N.next = temp.next
temp.next = N
```

Dry Run:

10 -> 20 -> 30 -> 40 -> Null

Final List:

10 -> 20 -> 25 -> 30 -> 40 -> Null

Deletion from Beginning

Example: Delete first node from

10 -> 20 -> 30

Pseudocode:

```
temp = head
head = head.next
delete temp
```

Dry Run:

Head -> 10 -> 20 -> 30 -> Null

Final List:

Head -> 20 -> 30 -> Null

Delete from End

Example: Delete last node from

10 -> 20 -> 30 -> 40

Pseudocode:

```
temp = head
```

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```
while temp.next.next ≠ NULL
    Temp = Temp.next
delete Temp.next
    Temp.next = NULL
```

Dry Run:

10 -> 20 -> 30 -> 40 -> Null

Final List:

10 -> 20 -> 30 -> Null

Delete from a specific position

Example: Delete node at position 2 from
10 -> 20 -> 30 -> 40

Pseudocode:

```
temp = head
for i = 1 to position-2
    temp = Temp.next
deleteNode = Temp.next
Temp.next = deleteNode.next
delete deleteNode
```

Dry Run:

10 -> 20 -> 30 -> 40 -> Null

Final List:

10 -> 30 -> 40 -> Null